ML Sports Betting: Project Proposal

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Project Summary

We will build a calibrated machine-learning betting system for the NBA and English Premier League soccer, and compare odds across several betting platforms to find both value bets and arbitrage opportunities. Our methodology follows Walsh and Joshi (2024), who argue that a betting model should be selected on **calibration** rather than accuracy, since profit depends on how close predicted probabilities are to the true probabilities.

Motivation

A bet has positive expected value only when the true probability of an outcome exceeds the probability implied by the odds (1/odds). Different platforms price the same game differently, so the best available odds for each outcome often sit on different platforms. This creates two opportunities we will study:

- **Value betting:** our calibrated model's probability beats the best available implied probability.
- **Arbitrage:** the best odds across platforms satisfy 1/O(home) + 1/O(away) [+ 1/O(draw)] < 1, guaranteeing a profit regardless of outcome.

Methodology

We keep the paper's pipeline and apply it to as many sports as possible (depending on data availability).

1. **Features:** season-to-date averages of each team's box-score out-performance versus opponents, plus prior-season win percentage, with Kolmogorov–Smirnov tests to drop features showing covariate shift.
2. **Two-branch model selection:** Spearman correlation filter, sequential forward feature selection, and Bayesian hyperparameter tuning (BO-TPE) over logistic regression, random forest, SVM and MLP. One branch maximizes accuracy and the other minimizes classwise expected calibration error (ECE). For soccer, classwise-ECE handles the three outcomes (home, draw, away).
3. **Chronological splits:** train, validate and test on past seasons only, then simulate betting on a held-out season.
4. **Multi-platform odds:** for every game, take the best odds per outcome across platforms, flag arbitrage opportunities, and flag value bets from each model.
5. **Staking and evaluation:** fixed-stake and fractional-Kelly betting from a fixed starting bankroll, comparing calibration-driven and accuracy-driven systems by ROI, with and without line shopping across platforms.

Data Sources

- Yet to be explored. Suggested platforms for data: Kaggle, sportsbookreviewsonline.com and Las Vegas sportsbook Westgate.

Reference

Walsh, C. & Joshi, A. (2024). Machine learning for sports betting: Should model selection be based on accuracy or calibration? *Machine Learning with Applications*, 16, 100539. https://doi.org/10.1016/j.mlwa.2024.100539